RAJALAKSHMI ENGINEERING COLLEGE

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M2

Started on	Thursday, 6 August 2026, 11:02 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:08 AM
Time taken	6 mins 26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Converttthefollowing algorithm into a program and find its time complexity using counter method.

```
Factor(num){
{
   for (i = 1; i <= num;++i)
   {
    if(num%i==0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int count=0;
void Factor(int num){
    for(int i=1;i<=num;++i){
        count++;
        count++;
        if(num%i==0){
            count++;
        //    printf("%d ",i);
        }
    }
    count++;
}
int main(){
    int num;
    scanf("%d",&num);
    Factor(num);
    printf("%d",count);
}
```

	Input	Expected	Got	
				12	31	31				
	25	54	54	
	4	12	12	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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